Supplementary Information

Architecture-dependent effects of imbalance correction in melanoma classification: a controlled audit with fairness, calibration, and explanation analysis

S1 Per-seed Stage 2 metrics

Table 1 reports every Stage 2 run. The main text reports seed-level means and 95% t intervals.

Table 1: Per-seed Stage 2 test metrics at the validation-selected 95% specificity threshold.

Architecture	Condition	Seed	Sensitivity	Specificity	AUC
ResNet-50	CE	1	0.608	0.944	0.896
ResNet-50	CE	2	0.527	0.961	0.901
ResNet-50	CE	3	0.575	0.949	0.895
ResNet-50	CE+strong	1	0.602	0.936	0.910
ResNet-50	CE+strong	2	0.554	0.943	0.890
ResNet-50	CE+strong	3	0.548	0.948	0.897
ResNet-50	Focal+CB	1	0.435	0.943	0.881
ResNet-50	Focal+CB	2	0.398	0.955	0.885
ResNet-50	Focal+CB	3	0.527	0.942	0.887
EfficientNet-B0	CE	1	0.543	0.956	0.917
EfficientNet-B0	CE	2	0.527	0.949	0.905
EfficientNet-B0	CE	3	0.597	0.955	0.920
EfficientNet-B0	CE+strong	1	0.570	0.952	0.909
EfficientNet-B0	CE+strong	2	0.624	0.946	0.922
EfficientNet-B0	CE+strong	3	0.495	0.957	0.905
EfficientNet-B0	Focal+CB	1	0.522	0.947	0.884
EfficientNet-B0	Focal+CB	2	0.392	0.946	0.882
EfficientNet-B0	Focal+CB	3	0.516	0.955	0.907
ViT-S/16	CE	1	0.349	0.949	0.850
ViT-S/16	CE	2	0.430	0.937	0.842
ViT-S/16	CE	3	0.419	0.938	0.870
ViT-S/16	CE+strong	1	0.349	0.942	0.853
ViT-S/16	CE+strong	2	0.516	0.942	0.877
ViT-S/16	CE+strong	3	0.441	0.938	0.868
ViT-S/16	Focal+CB	1	0.505	0.944	0.878
ViT-S/16	Focal+CB	2	0.462	0.928	0.860
ViT-S/16	Focal+CB	3	0.478	0.948	0.880

S2 XAI error-type mass

Table 2 reports attribution mass for false negatives and false positives with Grad-CAM targeting the true class. The values do not separate consistently across architectures.

Table 2: Attribution mass by error type for the true-class Grad-CAM analysis. Values are means across images and seeds.

Architecture	Error type	Images	Attribution mass
EfficientNet-B0	FN	23	0.374
EfficientNet-B0	FP	31	0.260
ResNet-50	FN	22	0.346
ResNet-50	FP	25	0.349
ViT-S/16	FN	30	0.420
ViT-S/16	FP	31	0.344

S3 Training and ROC figures

Figures 1 and 2 show the training curves and test ROC curve for the median seed of ResNet-50 with the CE baseline. These figures were moved out of the main text because they are illustrative and were generated at 150 dpi.

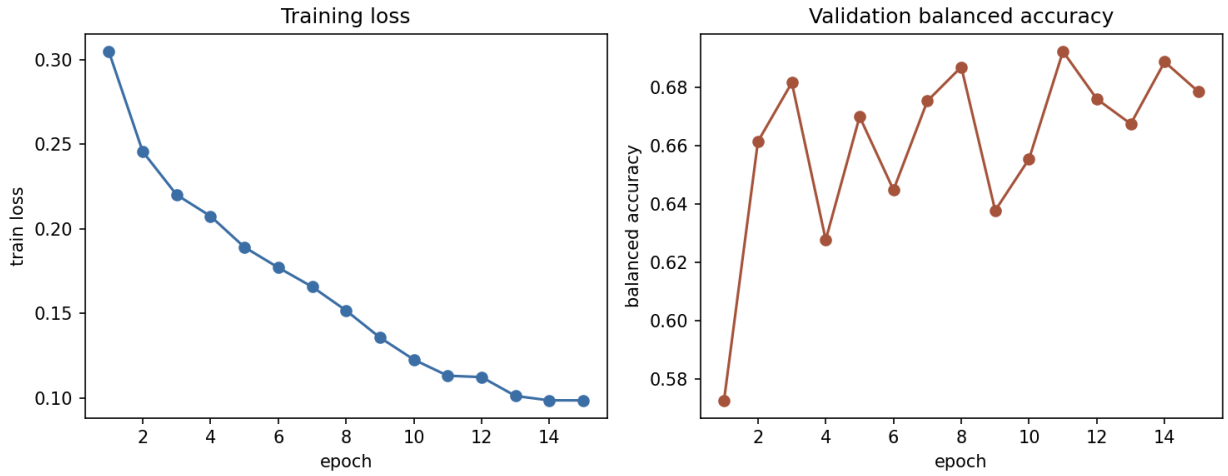


Figure 1: Training loss and validation balanced accuracy for the median seed of ResNet-50 with the CE baseline.

S4 Data, code, and additional results

The lesion-aware split is released as `results/split_ham10000_lesion_seed42.csv`. Per-sample test predictions for all 27 runs are in `results/predictions/`. The full adjusted-model coefficients, dark-skin AUC comparisons, calibration summaries, and XAI metrics are available from the corresponding author upon reasonable request. The repository link is withheld for double-blind review.

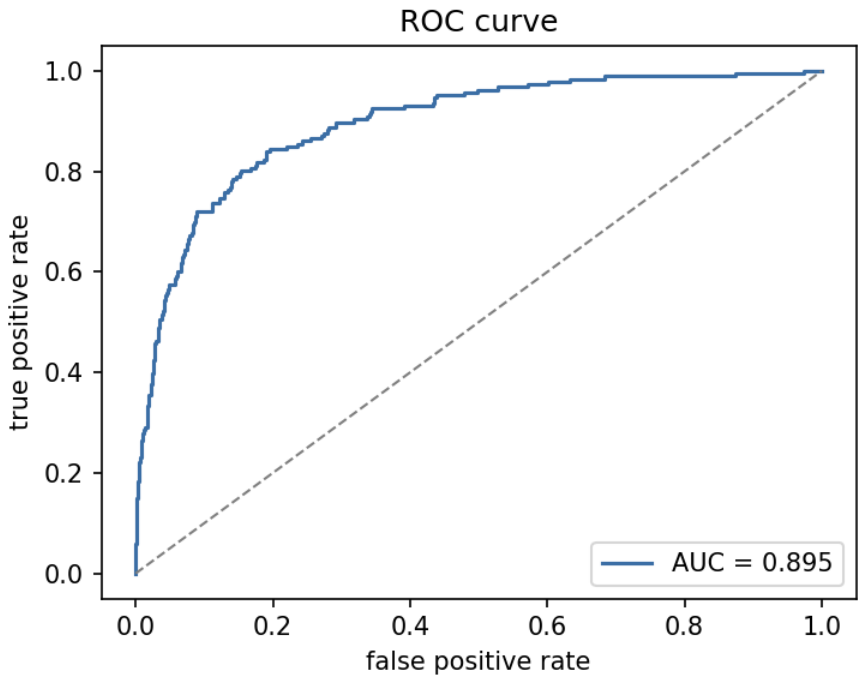


Figure 2: Test ROC curve for the median seed of ResNet-50 with the CE baseline. The operating threshold is selected on validation at 95% specificity.